

# 18-427/727: Board-Level RF Systems for IoT

## LAB03: Patch Antenna Characterization

**Writeup Due:** Monday, October 27th, 2025

### How to submit labs:

Create a word document or PDF that answers the questions presented below. Include any screenshots or figures as requested. **Please provide a descriptive caption for each figure!**

|           |    |    |    |       |
|-----------|----|----|----|-------|
| Question: | 1  | 2  | 3  | Total |
| Points:   | 20 | 30 | 50 | 100   |
| Score:    |    |    |    |       |

# Lab Outline

This lab is a followup to Lab 2, where you designed your own patch antenna at a specific center frequency. The TAs have milled out the boards, and now you will get a chance to take some measurements of the antenna to plot its radiation pattern. We will be using the VNA to take measurements by connecting a receive antenna to Port 2 and your patch antenna to Port 1, then measuring the  $S_{21}$  value at the patch antenna's center frequency as the antenna rotates 360 degrees. This will result in a plot of the radiation pattern.

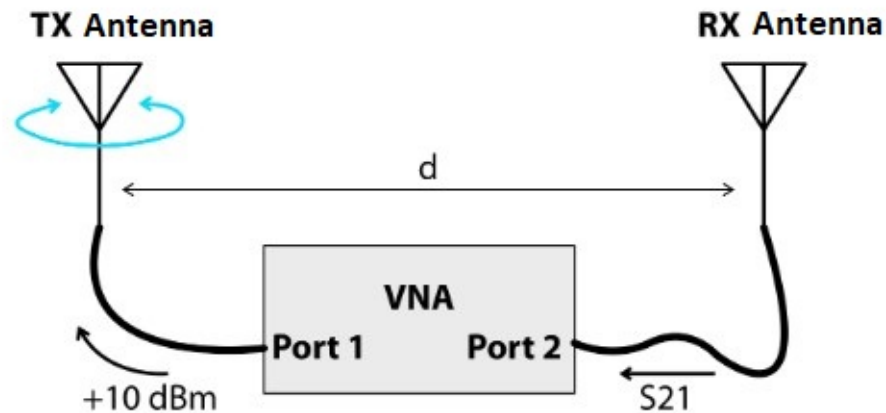


Figure 1: Diagram of the test setup. The VNA provides -10 dBm to a transmit patch antenna under test from Port 1, then the RX whip some distance away receives the radiated signal and passes it to Port 2. Measuring  $S_{21}$  as the TX patch rotates will allow us to visualize its pattern.

## 1. Setup

- Soldering the SMA connector
- Setting up the experiment

## 2. Measurements

- Calibrating the VNA
- Observing the power absorption
- Running the Python script
- Visualizing the data

## 3. Analysis

## Equipment Required

- Your team's patch antenna and an SMA connector - you will receive these from the TAs. Please note that if we are unable to provide you with your patch antennas, then we will just run the experiment with an existing patch antenna from Lab 1.
- Anechoic chamber and equipment setup - this lab will be run in REH224, which has an anechoic chamber. This lab space can be used to characterize antennas in a low-noise environment. You will meet with the TAs for a 15-minute time slot to run your experiment; they will send you the data afterwards for analysis. The equipment setup in the lab includes a horn antenna, a LibreVNA, and a Raspberry Pi.

## 1. Setup

The TAs have milled out each team's antenna. Ask the TA for your team's antenna board, along with an SMA connector. You will get some experience in soldering the connector to the board so that you may take antenna measurements.

Below is a photo of the patch antenna from Lab 1:

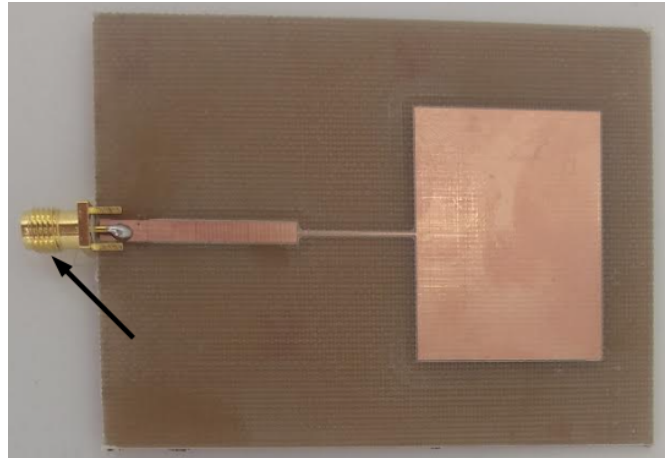


Figure 2: Patch antenna with edge-mounted SMA connector. It's your job to solder the connector on!

Take note of how the solder has been melted and dispersed across the copper surface. The solder should not form a ball on top of the copper; if this happens, then you may not get a good electrical connection! Because the large traces of copper wick heat away from the solder, you may need to increase the temperature of the iron (above 750°F) to get good solder joints. Careful - that copper ground plane will retain heat for some time! Also, note that for these SMA connectors, the leg in the center is what connects to the RF signal; all other legs are grounded. Be careful not to short the center leg to any of the other legs:

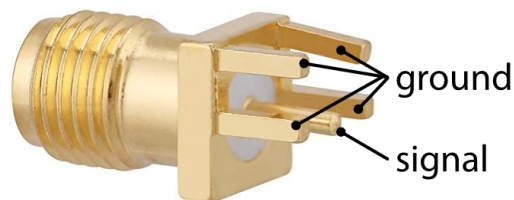


Figure 3: Edge-mounted SMA connector, legs annotated.

Once you are done, perform a continuity test with a digital multimeter to ensure that nothing has been shorted. You should put one probe on the antenna's feedline, and the other probe on the ground plane; it should be high impedance since they are electrically isolated. After this, move one probe from the ground plane to the inside of the SMA connector (where the RF cable is inserted); you should see electrical continuity. **Take photos of the front and back of your antenna with the SMA connector soldered on.** Please make sure to complete this before your scheduled lab session in REH224!

Now, you are prepared to characterize your patch antenna. The TAs have set up the anechoic chamber in REH224 so that you may take measurements of your antenna. The system consists of a LibreVNA and a Raspberry Pi, which is running the LibreVNA GUI and scripts to collect data. You can see what the inside of the anechoic chamber looks like below; the horn antenna is located at the far end of the chamber, adjacent to the light:



Figure 4: Anechoic chamber in REH224.

Mount your patch antenna to the SMA connector for the device-under-test. The TAs have already done two-port calibration for the setup, so you should be all set to run the experiment. If you're curious about the setup in the anechoic chamber, please see [this repository](#) for more information.

10 pts

**1.1** Include a photo of the front and back of your antenna with the SMA connector soldered on. (Note: if we are unable to provide you with your physical antenna, then please disregard the above instructions; you may just upload screenshots of your PCB in KiCAD to receive credit for this part.)

10 pts

**1.2** The patch antenna and the horn antenna are several meters apart. Why do we want the antennas to be some minimum distance apart when measuring patterns?

## 2. Measurements

We are now ready to start taking measurements. First, we will make a simple observation of the antenna setup. The LibreVNA GUI in REH224 is set up so that it shows the S11 and S21 values on a plot against frequency.

With the receive antenna disconnected from the VNA (on Port 2), **take note of which frequency gives the minimum value of S11**; this is your patch antenna's center frequency. Once you have this center frequency, reconnect the receive antenna to Port 2.

We can control the device-under-test using a motor controller which rotates the table that the antenna is attached to. The TAs will help you position the patch antenna's broad side to face the receive antenna. Then, they will rotate it slowly by 90 degrees so its thin side is now facing the receive antenna. **Take a screenshot of your S21 plot at 0 degrees and at 90 degrees.**

Now, we can collect some data that will tell us a bit more information about the radiation pattern. View the following files from Canvas:

- libreVNA.py - contains helper functions that let us interact with the VNA via SCPI
- radiation\_pattern.py - contains the main functions used to get data and plot the radiation pattern

The VNA can be communicated with using SCPI commands, which lets you control the device from a script (see [here](#) for more information on SCPI). This will be very helpful for us, since we are going to be taking a number of measurements, which would be quite tedious to do manually via the GUI. The main command to start the experiment is: **python3 anechoic\_chamber.py**

When we run **anechoic\_chamber.py**, the Raspberry Pi will attempt to connect to the VNA; if the VNA is connected, then it should print a message indicating so. With the center frequency provided in the **.config** file, the script will now begin taking S21 measurements at that center frequency. After each measurement, it will print the S21 as a complex number and then send a command to rotate the motor by a fixed number of degrees. Once the motion is complete, the VNA takes another measurement. It takes several seconds for the VNA to complete each measurement, so the entire process may take several minutes.

Once it has taken 96 measurements (one every 3.75 degrees), the script will now take each measurement and turn it into a normalized gain (in dB). Then, it will produce a polar plot of these values! Since we did not take 360 measurements, the polar plot will not be totally smooth. But it should give you a good idea of the antenna's radiation pattern - in particular, the antenna's directivity! **Take a screenshot of this plot.**

10 pts

**2.1** Share your screenshot of the S11 sweep from 2 GHz to 3 GHz. What was your antenna's target center frequency, and what was the actual frequency measured? Why do you think there was a discrepancy, if any?

10 pts

**2.2** Share your screenshots of the S21 sweep from 2GHz to 3GHz at both 0 degrees and 90 degrees. What did you notice about the S21 plot as you rotated the transmit antenna slowly from 0 degrees to 90 degrees (i.e. facing the receive antenna directly compared to facing perpendicular)?

10 pts

**2.3** Share the screenshot of your radiation pattern plot. What is the approximate 3dB beamwidth of your antenna's main lobe?

### 3. Analysis

Lastly, we will ask some additional questions that will help you reflect on this lab:

5 pts

**3.1** In previous semesters, we ran this experiment in the TechSpark classroom by constructing the same setup shown in Fig. 1 using a whip antenna for the RX antenna. Why would this give suboptimal experimental results compared to our setup in the anechoic chamber?

10 pts

**3.2** In what direction did your antenna have maximum gain? Where did you see any nulls in the pattern? Is this what you expected for a patch antenna?

10 pts

**3.3** You will use the Friis transmission formula to answer this and the next question:

$$P_R = P_T G_T G_R \left( \frac{\lambda}{4\pi d} \right)^2$$

What is the approximate path loss in dB for your measurement RF link in dB?

10 pts

**3.4** Using your measurements, please calculate the combined antenna gain  $G_T G_R$  for your measurement RF link. Remember that the output power from the VNA is -10 dBm.

5 pts

**3.5** In this lab, you are measuring a relative antenna pattern, normalized to the maximum. What is preventing you from presenting your data in units of dBi or dBd?

10 pts

**3.6** How could you improve your antenna design for different metrics (directivity, efficiency, etc.)?

# Reflections

Please complete our [post-lab survey](#) for 5 points of extra credit! Attach a photo of the form completion page to your lab submission.